

ARXIV 2504.05118 · BYTEDANCE SEED

VAPO

Value-Model-Based RL for Advanced Reasoning

A reinforcement learning framework that systematically addresses value-model bias, heterogeneous sequence lengths, and reward sparsity — achieving state-of-the-art mathematical reasoning on a single base model with no SFT data.

QWEN2.5-72B

NO SFT DATA

5K STEPS

60.4

AIME 2024 AVG@32

DAPO

50

DeepSeek-R1-Zero-32B

47

+10.4 pts above DAPO

ByteDance Seed · 2025

Why Value Models? Finer Credit Assignment for Long Chain-of-Thought Reasoning

VALUE-MODEL-FREE (GRPO / DAPO)

Trajectory-Level Advantage

- Advantage computed from final reward, applied uniformly to all tokens in trajectory
- No per-step credit distinction — all tokens weighted equally
- High variance: single Monte Carlo estimate per trajectory
- Subtle step-level errors are indistinguishable from correct steps
- Dominated LLM RL training prior to VAPO

VALUE-MODEL-BASED (VAPO)

Token-Level Credit Assignment

- **Precise attribution:** traces each token's impact on returns via bootstrapped value estimates
- **Lower variance:** per-token value estimates vs. Monte Carlo trajectory average
- **Generalization:** trained value model shares information across samples
- **Critical for multi-step:** step-level errors in long CoT cause catastrophic failures — granular credit is essential
- Enables VAPO to learn which reasoning steps are genuinely useful

Three Challenges Plaguing Value-Model-Based RL in Long Chain-of-Thought Tasks

01

Value Model Bias

Long trajectories with bootstrapped learning introduce systematic bias into value estimates. Errors compound across many reasoning steps, making value estimates unreliable at the start of training.

HIGH PRIORITY

02

Heterogeneous Sequence Lengths

Reasoning traces span highly variable lengths — from short confirmations to multi-page derivations. A single fixed bias-variance tradeoff (λ) cannot be optimal for all response lengths.

NOVEL CHALLENGE

03

Reward Signal Sparsity

Only the final `<eos>` token receives a reward signal. In long CoT, this sparsity is extreme — thousands of tokens derive signal from a single endpoint, weakening the learning gradient.

EXACERBATED BY COT

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL

CHALLENGES ADDRESSED

- Value Model Bias
- Heterogeneous Lengths
- Reward Sparsity

VAPO COMPONENTS

Value-Pretraining

Pre-trains the value model before RL to reduce initial bias from bootstrapped estimates on long trajectories.

BIAS

Length-Adaptive GAE ★

Dynamically adjusts λ per response length. Short $\rightarrow$ low λ (low variance). Long $\rightarrow$ high λ (low bias).

NOVEL · LENGTH

Positive Example LM Loss

Self-imitation learning on correct trajectories to reinforce high-quality reasoning chains.

SPARSITY

Decoupled-GAE

Separates GAE computation from policy for more robust value estimation. Adapted from VC-PPO.

BIAS

Token-level Loss

Normalizes loss at token level instead of sequence level, preventing length bias in gradient updates.

SPARSITY

Clip-Higher + Group-Sampling

Asymmetric clipping (DAPO) for exploration; multi-response sampling (GRPO) for baseline estimation.

MULTI



Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths

In standard GAE, the λ parameter controls the bias-variance tradeoff uniformly. But long-CoT reasoning produces **wildly heterogeneous** response lengths — a single fixed λ is suboptimal for both short and long sequences simultaneously.

- $\lambda \downarrow$ **Short responses** → lower λ prioritizes lower variance. Fewer bootstrapping steps mean bias is already low; reducing λ controls noise.
- $\lambda \uparrow$ **Long responses** → higher λ prioritizes lower bias. Extended reasoning chains accumulate bootstrapping errors; higher λ leverages more returns to correct this.
- ★ Novel contribution — no prior work in LLM RL addresses length-dependent GAE parameterization. Contributes **+15 points** in ablation (45 → 60).



Short Response

Few reasoning steps · low bootstrapping error

$\lambda \downarrow$ low

: Dynamic $\lambda(\text{length})$



Long Response

Extended reasoning chain · accumulated bias

$\lambda \uparrow$ high

Ablation result: Removing Length-Adaptive GAE drops AIME 2024 score from **60** → **45** — a 15-point contribution that cannot be recovered by any other component. This challenge is *unique* to long-CoT reasoning tasks.

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero

AIME 2024 AVG@32 · QWEN2.5-32B BASE · NO SFT DATA



+10 pts over DAPO
Previous SOTA on same Qwen2.5-32B base

+13 pts over DeepSeek-R1-Zero
Both use same base model, no SFT

+55 pts over Vanilla PPO
Demonstrates value of VAPO's full framework

All comparisons on Qwen2.5-32B pre-trained base model with no supervised fine-tuning (SFT) data. VAPO reaches 60.4 within ~5,000 training steps.

VAPO Reaches SOTA in 5K Steps — Consistent Lead and Longer Reasoning Chains

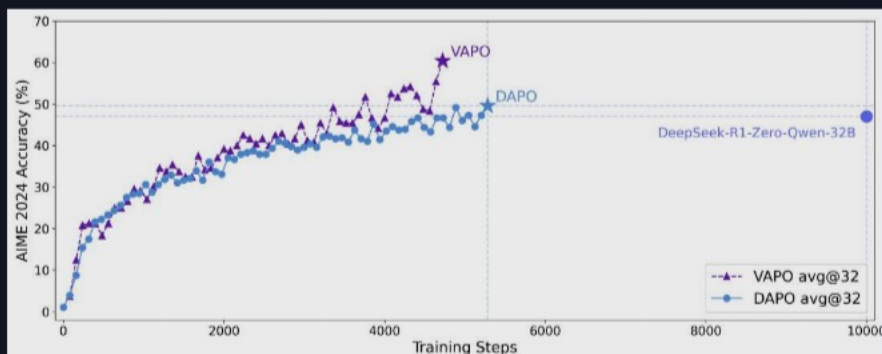


Fig. 2 — Accuracy vs. training steps. VAPO consistently leads DAPO throughout training, reaching ~60% at step 4,500. DAPO plateaus near 50% at step 5,500.



Fig. 3 — Mean response length over training. VAPO generates ~6,000-token responses vs. ~4,500 for DAPO — evidence of more elaborate reasoning chains.

~5K

Steps to SOTA

0

Training crashes (multiple runs)

~6K

Avg. response tokens (VAPO)

Ablation: Every Component Matters – Value-Pretraining Alone Adds 49 Points

CONFIGURATION	AIME 2024	Δ VS VAPO
VAPO (Full)	60	–
w/o Group-Sampling	55	–5
w/o Positive Example LM Loss	54	–6
w/o Token-level Loss	53	–7
w/o Clip-Higher	46	–14
w/o Length-adaptive GAE	45	–15
w/o Decoupled-GAE	33	–27
w/o Value-Pretraining CRITICAL	11	–49

- 49 Value-Pretraining** is the single most critical component. Without it, value-model-based RL barely works (11 vs. 60) – the model cannot bootstrap from biased values.
- 27 Decoupled-GAE** is the second most impactful. Separating GAE from the policy update is essential for stable value-based learning (33 vs. 60).
- 15 Length-Adaptive GAE** contributes 15 points – a novel technique essential for heterogeneous sequence lengths (45 vs. 60).
- 7** Remaining components (Token-level Loss, Positive LM Loss, Group-Sampling) each contribute incrementally, validating the full integrated design.

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

01 / Most Important

Value pretraining is load-bearing

Without it, value-model-based RL achieves only 11 on AIME 2024. Addressing initial value bias is a prerequisite, not an optimization.

02

Value-model-based > value-free for complex reasoning

When all three challenges (bias, heterogeneous lengths, reward sparsity) are addressed, value models provide token-level credit that value-free methods cannot match.

03

Length-Adaptive GAE is a novel, critical technique

Dynamically adjusting λ based on response length adds 15 points and addresses a challenge unique to long-CoT reasoning — applicable to any value-model RL system.

04

Training efficiency and stability achieved

VAPo reaches SOTA in ~5,000 steps with zero training crashes across multiple independent runs — competitive with, and often faster than, value-free methods.

05 / Implications

Roadmap for future reasoning model development

VAPo demonstrates that systematic engineering — not just scale — unlocks value-model-based RL for long-CoT. The three-challenge framework (bias · length heterogeneity · reward sparsity) provides a blueprint for future work on mathematical and scientific reasoning models.